Contest-aligned End-to-End InSAR + Deep Learning Pipeline for the IEEE GRSS 2026 Data Fusion Contest

Executive summary

The contest dataset is a rare "realistic commercial-stack" testbed: about **1582 unique collects**, enabling **17000+ possible interferometric pairs**, with some AOIs having **50+** acquisitions over a year, and substantial variability in **mode, incidence angle, look direction, and orbital geometry**. □cite□turn9view0□turn19view0□ The evaluation is committee-based and rewards **soundness, originality, insight, effective dataset use, scalability (optional), and reproducibility with open code and strong visuals**. □cite□turn1view0□turn19view0□ Submission deadline is **April 06, 2026 (23:59 AoE)**. □cite□turn19view0□

A winning approach must be explicitly **stack-aware** and **geometry-aware**. The most defensible, contest-native contribution is a pipeline that:

1. **builds a pair-graph** (nodes = collects, edges = candidate pairs) and selects a high-value subset using metadata and learned quality prediction,
2. computes baseline interferograms/unwrapping/DEM/time-series products using standard InSAR,
3. trains a deep model to **improve interferometric products** (denoised complex interferogram + per-pixel uncertainty) using **self-supervision + closure + temporal-consistency losses**,
4. integrates uncertainty into unwrapping and inversion (weighted DEM/time-series), and
5. reports improvements with hard, physics-linked metrics: **triplet closure error**, **unwrap success**, **temporal residuals**, **percent usable pairs**, and **DEM NMAD**. □cite□turn16search0□turn15search8□turn7view0□turn19view0□

Key "win targets" you should aim to demonstrate (vs the strongest classical baseline you can run):

- **Median closure error ↓ 30–50%** on stable pixels and across multiple AOIs/modes. □cite□turn16search0□turn16search13□
- **Unwrap success rate ↑ +15–25 percentage points** at fixed quality gate. □cite□turn15search8□
- **Percent usable pairs ↑ +20–40%** (at fixed thresholds on coherence/closure/unwrap). □cite□turn9view0□turn19view0□
- **DEM NMAD ↓ 15–25%** (where baselines allow DEM sensitivity and where an external reference DEM is used only for evaluation). □cite□turn7view0□turn7view1□
- **Temporal consistency residual ↓ ≥20%** (stack inversion residuals). □cite□turn16search1□turn16search21□

Assumptions (explicit where unspecified):

- Public S3 access with no authentication (contest page states direct retrieval from S3 without authentication; AWS registry shows `--no-sign-request`). □cite□turn1view0□turn13view0□
- The STAC is a **static catalog** (not a hosted STAC API) unless you stand up a local index/API; the STAC root is a `Catalog` JSON and does not advertise a `/search` endpoint. □cite□turn5view0□turn0search17□turn0search5□
- Full dataset download is optional; competitive entries can use a carefully selected subset if the method is demonstrably scalable and generalizes across geography and acquisition diversity. □cite□turn19view0□turn9view0□
- External reference DEM use is optional and disclosed; used only for evaluation and not as a processing "crutch." □cite□turn19view0□turn7view0□

Contest framing, objectives, and success criteria

Contest constraints that shape your pipeline

1. The contest is single-stage: you submit a **max 4-page paper** (excluding references), with figures, and optional supplemental material. □cite□turn19view0□turn1view0□
2. Code must be **publicly accessible** (e.g., GitHub), and reproducibility is explicitly scored. □cite□turn1view0□turn13view0□
3. Winners are not selected by a numeric leaderboard but by committee judgment across technical dimensions. □cite□turn1view0□turn19view0□
4. Critical calendar dates: data released Feb 04; submissions due Apr 06; winners Apr 13; internal paper deadline Apr 20; final paper Apr 28. □cite□turn19view0□

Objectives (what your project should claim)

Define objectives in a way that mirrors the official evaluation criteria.

1. **Objective O1: Increase stack usability** under geometry diversity by improving interferogram quality and unwrapping reliability, quantified by closure and unwrap metrics. □cite□turn1view0□turn16search0□
2. **Objective O2: Improve downstream products**, including DEM (where baselines permit) and time-series consistency, using uncertainty-weighted inversion. □cite□turn9view0□turn7view0□turn16search1□
3. **Objective O3: Produce interpretable "temporal storytelling" artifacts** (maps + timelines + pair graphs) that reveal patterns/events and clearly show why your method helps. □cite□turn1view0□turn9view0□
4. **Objective O4: Reproducibility**: one-command pipeline from download → results figures, plus trained model weights and configs. □cite□turn1view0□

Success criteria and metric definitions table (requested)

Use the following table as the project's metric hub; implement all metrics and publish scripts.

Metric	Definition (explicit)	How to compute	Why it's load-bearing		
Triplet closure error	For three acquisitions (i,j,k), compute (c_(p)=\mathrm(\phi_(p)+\phi_(p)-\phi_(p))). Aggregate as median(_)((	c_(p)	)) and then median over triplets. Triplet closure is widely used for diagnosing inconsistency and unwrapping errors. □cite□turn16search0□turn16search13□	Build triplets from your selected graph; compute on stable mask; report distribution and quantiles.	Closur denois consis □cite□
Unwrap success rate	Define pass/fail per interferogram: (i) unwrapped connected component covers ≥90% of stable mask; (ii) triplet closure quantile gate (e.g., 95% < 1 rad) for triplets containing that edge. Use SNAPHU baseline for reproducibility. □cite□turn15search8□turn16search0□	Run unwrapping; compute connected component coverage; compute closure gates.	Unwrapping is a principal failure mode; improving it is directly valuable and contest-relevant. □cite□turn15search8□turn1view0□		
Percent usable pairs	% of candidate edges passing: median coherence > (c_0) (e.g., 0.35), unwrap success, closure gate. Coherence is standard interferometric quality output. □cite□turn0search2□turn9view0□	Define gates; compute at fixed thresholds across AOIs.	The dataset allows 17000+ pairs; scalability means selecting more usable pairs. □cite□turn9view0□		
DEM NMAD	Let DEM errors be (e=h_{\text}-h_{\text}). NMAD (=1.4826\cdot\mathrm(	e-\mathrm(e)	)). □cite□turn7view0□turn7view1□	Compare to chosen reference DEM on evaluation AOIs; compute NMAD on stable terrain mask.	NMAD DEM e □cite□
Temporal consistency residual	Stack inversion residual: solve (Ax\approx\phi) (SBAS-like); compute (|W(Ax-\phi)|_2) or per-edge residual quantiles. Phase consistency is explicitly important in SBAS variants. □cite□turn16search1□turn16search21□	Build design matrix; use weights from uncertainty/coherence; report residual maps and statistics.	Proves improvements survive time-series linking, not just single interferogram cosmetics. □cite□turn16search1□turn16search21□		

Recommended evaluation thresholds (targets, not guaranteed outcomes):

- closure error median ↓ ≥30%
- unwrap success ↑ ≥15 percentage points
- usable pairs ↑ ≥25%
- temporal residual ↓ ≥20%
- DEM NMAD ↓ ≥15% □cite□turn16search0□turn7view0□turn15search8□

Data access, STAC querying, and compute/storage planning

Data access: what is officially provided

1. The contest states the data are in a **public AWS S3 bucket**, accessible via a published STAC catalog and downloadable **without authentication**. □cite□turn1view0□turn19view0□
2. AWS registry listing provides the bucket ARN, region, and no-auth CLI example:

- S3 bucket: `s3://capella-open-data/data/` region `us-west-2`
- Example: `aws s3 ls --no-sign-request s3://capella-open-data/data/` □cite□turn13view0□

3. The Capella Open Data STAC root catalog (`stac/catalog.json`) contains a child link titled **"IEEE Data Contest 2026"** pointing to `capella-open-data-ieee-data-contest/collection.json`. □cite□turn5view0□
4. The contest page provides a CSV with per-scene/pair metadata headers including mode, look direction, incidence values and differences. □cite□turn9view0□turn10view0□

Static STAC vs STAC API (important tooling truth)

- **Static STAC**: a set of JSON files linked via `root/child/item` relations (what the S3-hosted Capella catalog is). □cite□turn5view0□turn0search5□

- **STAC API**: a dynamic, searchable REST service that implements `/search` and other endpoints; it is defined by OGC and STAC API specs. □cite□turn0search17□turn0search1□

Because the contest distribution is a static catalog on S3, `pystac` is the correct primary library. `pystac-client` becomes relevant if you build a local search index (recommended for speed and reproducibility). □cite□turn0search5□turn0search17□

Step-by-step data access methodology (numbered hierarchy)

Tooling installation

1. Install core tools: 1.1) AWS CLI v2
   1.2) Python packages: `boto3`, `pystac`, `rasterio`, `capella-reader`, `numpy`, `pyproj`, `shapely`, `geopandas`, `xarray`, `dask`. □cite□turn13view0□turn2view2□
   1.3) Optional: `rio-stac` (helpful for some STAC workflows; treat as optional because ecosystems vary).

2. Install SAR/InSAR processing: 2.1) ISCE3 (preferred for Capella SLC given `capella-reader` includes an ISCE3 adapter). □cite□turn2view2□turn4view1□
   2.2) SNAP as an optional baseline/reference workflow (often easiest for coherence/ifg baselines, though best supported for ESA missions). □cite□turn0search2□
   2.3) GAMMA only if you already have a license; otherwise treat as non-open baseline.

No-auth S3 listing and download (illustrative commands)

3. Validate open bucket access:

```
# Illustrative
aws s3 ls --no-sign-request s3://capella-open-data/data/
```

□cite□turn13view0□

4. (Recommended) compute actual dataset size for your selected prefixes:

```
# Illustrative; use --recursive and --summarize to estimate bytes
aws s3 ls --no-sign-request --recursive --summarize s3://capella-open-data/data/ > s3_inventory.txt
```

This is the only honest way to size storage since the contest does not publish total bytes. □cite□turn13view0□turn19view0□

STAC crawl (static) and local indexing

5. Crawl the STAC root and locate the contest collection:

```
# Illustrative: static catalog crawl (pystac)
import pystac

root = pystac.Catalog.from_file(
    "https://capella-open-data.s3.us-west-2.amazonaws.com/stac/catalog.json"
)

# Locate the IEEE contest collection via child links
contest_href = None
for link in root.links:
    if link.rel == "child" and "ieee-data-contest" in (link.href or ""):
        contest_href = link.href
        break

contest = pystac.Collection.from_file(
    "https://capella-open-data.s3.us-west-2.amazonaws.com/stac/"
    "capella-open-data-ieee-data-contest/collection.json"
)
print(contest.id)
```

The existence of the contest collection link is explicit in the root catalog JSON. □cite□turn5view0□turn6view0□

6. Build a local index (highly recommended): 6.1) Traverse all items, write each item JSON to `data/stac_cache/items/<item_id>.json`.
   6.2) Extract key fields to `parquet`:
   - item_id, datetime, bbox, mode, look_direction, flight_direction, incidence, assets (hrefs, sizes).
     6.3) Optional: stand up a local STAC API (e.g., using a lightweight FastAPI wrapper) pointing to this index for fast search during experiments; document it as an enhancement, not required by contest.

Storage/compute sizing estimates and options tables (requested)

The only reliable size computation is inventorying S3 and summing file sizes. □cite□turn13view0□ Still, you can produce engineering estimates using product format facts:

- SLC pixel samples are **CInt16** (real Int16 + imaginary Int16) = **32 bits** per pixel sample. □cite□turn3view0□
- Capella delivers data as **Cloud-Optimized GeoTIFF + two JSON sidecars** (STAC + extended). □cite□turn3view2□

Approximate uncompressed SLC size: [\text \approx 4 \cdot H \cdot W] then adjust by GeoTIFF compression/tiling (unknown a priori). □cite□turn3view0□turn3view2□

Compute/storage budget options table (requested)

Tier	Intended scope	Typical SLC count	Pair count	Storage (local)	GPUs	Expected capability
Minimal	1–2 AOIs; proof-of-pipeline	40–120	300–1500	1–3 TB	1×24 GB	credible baseline + DL prototype
Competitive	3–6 AOIs; diversity across modes/incidence	200–600	3k–15k	6–20 TB	2×48 GB (or 4×24)	strong ablations + generalization claims
Ideal	broad-scale claim; many AOIs/modes	800–1582	10k–30k	20–60 TB (NVMe/RAID/object store)	4–8×80 GB	stack-scale benchmarks + best odds

The contest criteria explicitly reward scalability as optional but favorable, so “Competitive” tier is the practical minimum to claim cross-geometry generalization. □cite□turn1view0□turn19view0□

Preprocessing and InSAR baseline pipeline

Product formats and calibration facts you must implement correctly

1. SLC is single-look, focused SAR data in slant-range/azimuth geometry; pixel values are complex (Re/Im) stored as CInt16. □cite□turn3view0□
2. Capella TIFF+JSON bundle: one COG GeoTIFF + STAC JSON + extended JSON. □cite□turn3view2□
3. Radiometric conventions: SLC values relate to **Beta Nought**; GEO/GEC are radiometrically calibrated (Sigma Nought) and GEO uses terrain correction with a DEM. □cite□turn3view1□
4. A `scale_factor` (SC) is provided in extended metadata and is used to derive radar brightness from pixel values. □cite□turn3view1□
5. The `capella-reader` package parses metadata from TIFF tags/JSON and provides ISCE3 conversion utilities. □cite□turn2view2□

Step-by-step baseline InSAR workflow (with illustrative commands)

SLC ingestion and metadata parsing

1. For each selected collect: 1.1) Download `{product}.tif`, `{product}.json` (STAC), `{product}_extended.json` (extended). □cite□turn3view2□turn13view0□
   1.2) Parse metadata with `capella-reader`:

```
# Illustrative
from capella_reader import CapellaSLC
slc = CapellaSLC.from_file("path/to/slc.tif")
orbit = slc.adapted.isce3_orbit  # conceptual; actual adapter call per docs
```

□cite□turn2view2□

Pairwise coregistration (ISCE3-first recommendation)

2. Coregister SLC pairs: 2.1) Use Capella metadata (orbit, doppler) converted to ISCE3 structures via `capella-reader`. □cite□turn2view2□
   2.2) Perform coarse geometric alignment, then refine with correlation offsets.
   2.3) Save residual misregistration metrics; these become features for the usability predictor.

Illustrative wrapper command (your repo provides actual implementation):

```
# Illustrative
python run_coreg_isce3.py --ref ref_slc.tif --sec sec_slc.tif --out pair_dir/
```

Interferogram and coherence estimation

3. Form complex interferogram (I = S_\text\cdot S_\text^*) and estimate coherence in a window; coherence is standard and used as a quality indicator in common workflows. □cite□turn0search2□

Baseline filtering (non-DL denoisers)

4. Implement at least these phase/noise baselines: 4.1) **Goldstein adaptive filter**: classic interferogram filtering for geophysical applications. □cite□turn16search2□turn16search6□
   4.2) **NL-InSAR**: strong nonlocal estimator for interferograms. □cite□turn16search7□

4.3) **BM3D**: canonical nonlocal denoiser to include as “BM3D-like” baseline (applied carefully to periodic phase or to complex channels). □cite□turn15search17□turn15search13□

Unwrapping baseline

5. Unwrap with SNAPHU: 5.1) SNAPHU implements a statistical-cost network-flow MAP approach for phase unwrapping. □cite□turn15search8□
   5.2) Use a consistent coherence mask threshold across baselines (report mask percentage).

```
# Illustrative
snaphu -f snaphu.conf wrapped_phase.bin <width>
```

□cite□turn15search8□

Image group (helpful visual grounding)

□image_group□{"layout":"carousel","aspect_ratio":"16:9","query":["InSAR interferogram fringes coherence map example","SNAPHU phase unwrapping network flow diagram","Goldstein interferogram filtering example","NL-InSAR nonlocal interferogram estimation illustration"],"num_per_query":1}□

Pair-graph construction and edge scoring

Why the pair-graph is central for this contest

The contest explicitly highlights stack density and geometry diversity, and provides the CSV/metadata fields needed to build a pair selection mechanism. □cite□turn9view0□turn19view0□ A naive “all pairs under some baseline threshold” strategy will underperform and won’t look insightful.

Node and edge definitions

1. Node (v_i): a collect, with attributes:
   - `collect_id`, time, platform, mode, polarization,
   - look direction, flight direction,
   - incidence angle at scene center. □cite□turn9view0□turn11view0□
2. Edge (e_): candidate pair between collects (i) and (j), with attributes:
   - (\Delta t), (\Delta\theta_\text), (\Delta\theta_\text),
   - mode compatibility, look/flight consistency,
   - estimated perpendicular baseline (B_\perp),
   - SNR proxies (NESZ/peak noise fields, amplitude stability). □cite□turn9view0□turn3view1□turn2view2□

Computing perpendicular baseline (what to implement)

Assumption: (B_\perp) is not directly in the contest CSV; compute it from orbit/state vectors and radar geometry from extended metadata parsed by `capella-reader` plus ISCE3 utilities. □cite□turn2view2□turn3view0□

Explicit edge scoring formula (required)

Define a scalar score (Q_\in[0,1]) for pair usability prediction and graph selection:

[\begin Q_ &= \mathbf{1}[\text] \mathbf{1}[\text] \mathbf{1}[\text] \ &\quad \times \exp!\left(-\alpha_t \frac{\Delta t}{\tau_t}\right) \exp!\left(-\alpha_\text\frac{|\Delta\theta_\text|}{\tau_\text}\right) \exp!\left(-\alpha_g\frac{|\Delta\theta_\text|}{\tau_g}\right) \ &\quad \times \exp!\left(-\alpha_b\frac{|B_\perp|}{\tau_b}\right) \times \sigma(\beta_0+\beta_1,\widehat{\text}_) \end]

Where:

- (\widehat{\text}_) can include NESZ-like proxies and amplitude stability measures (Capella metadata includes NESZ fields and scale_factor per spec updates). □cite□turn2view0□turn3view1□
- (\Delta\theta_\text) and (\Delta\theta_\text) are explicitly defined in the contest CSV headers list. □cite□turn9view0□

Selection strategy:

1. For each AOI, select: 1.1) a **time-series subgraph** emphasizing small (\Delta t) and high (Q_), and
   1.2) a **DEM-sensitivity subgraph** emphasizing diverse (B_\perp) while keeping (Q_) above a minimum.
2. Cap total edges to a computational budget and document it.

Mermaid chart requested: pair-graph entity relationships

classDiagram class Collect { +collect_id: string +t: datetime +mode: string +look_dir: string +flight_dir: string +inc_deg: float } class PairEdge { +ref: collect_id +sec: collect_id +delta_t_days: float +delta_inc_deg: float +delta_graze_deg: float +Bperp_m: float +snr_proxy: float +Qij: float } class InSARProducts { +ifg_complex: raster +coherence: raster

+phi_wrapped: raster +phi_unwrapped: raster } Collect "1" --> "many" PairEdge : participates PairEdge "1" --> "1" InSARProducts : produces

Deep learning method design, training, and integration

Comprehensive literature review: DL in InSAR pipelines and stacks

This literature review is intentionally organized by *pipeline stage* because that is how your contest contribution will be judged (soundness + usefulness).

DL for interferogram/phase filtering and coherence estimation (CNNs)

- **DeepInSAR** proposes a CNN model for *phase filtering and coherence estimation* and motivates it as crucial for subsequent processing steps. □cite□turn0search0□turn0search16□
- DL filtering frameworks emphasize preserving informative phase while reducing noise to improve unwrapping robustness. □cite□turn10search2□turn0search20□
- **Φ-Net / Phi-Net** (DLR) is presented as a method for joint estimation of InSAR phase and coherence; the repository describes strategy and generalization aims. □cite□turn0search12□turn0search8□
- Classical baselines still matter: Goldstein filtering remains a widely cited operational baseline and must be included in comparisons. □cite□turn16search2□turn16search6□

DL for phase unwrapping (CNNs, GANs, transformers)

- Deep unwrapping networks exist (e.g., U-Net based) and often combine denoising and unwrapping to avoid error compounding. □cite□turn10search0□turn10search1□
- Transformer-based and “channel transformer” unwrapping networks are appearing (e.g., ResUCTransNet) and reflect the trend toward attention mechanisms for handling complex contexts. □cite□turn0search14□
- Transformer-based InSAR unwrapping under noisy conditions is explicitly being studied (recent reports/abstracts), supporting the relevance of attention for high-noise/heterogeneous settings. □cite□turn0search10□turn0search2□
- GAN-based approaches for interferometric phase reconstruction in decorrelated areas have been proposed, including multi-stage designs that reconnect fringes and then predict phase. □cite□turn0search5□turn0search21□

Caution: at least one GAN unwrapping paper in the search results is marked **retracted**; do not use it as a primary pillar. □cite□turn0search1□

DL and self-supervision in SAR despeckling (Noise2Noise, VAEs, diffusion)

- **Noise2Noise** establishes that learning from noisy–noisy pairs can work under common circumstances, making it a core conceptual tool for self-supervised training. □cite□turn15search2□
- Self-supervised SAR despeckling leveraging multi-temporal series is an established idea (e.g., SAR2SAR). □cite□turn10search7□
- VAEs/autoencoders are used in SAR representation learning and can jointly handle compression and despeckling (evidence that VAEs are relevant baselines/alternatives). □cite□turn1search2□
- Diffusion models are now directly applied to SAR despeckling (e.g., SAR-DDPM) and diffusion+transformer hybrids are reported. □cite□turn0search3□turn0search7□

Closure/temporal-consistency and stack-aware metrics (InSAR-specific)

- Triplet phase closure is critical as a quality indicator and for identifying unwrapping errors in networks. □cite□turn16search0□turn16search24□
- Closure and multilooking can introduce non-zero closure phase biases; closure needs careful handling in dense multilooked time series. □cite□turn16search13□turn16search5□
- SBAS variants and recent work emphasize phase consistency and temporal coherence metrics. □cite□turn16search1□turn16search21□

DL for DEM improvement and errors (scope carefully)

Your contest claim should be: “DL improves InSAR products including DEM *where geometry permits*,” not “DL always fixes DEM.”

- Multi-baseline InSAR DEM reconstruction is improved by combining multiple interferograms; modern discussions include methods that can reduce reliance on single-baseline unwrapping. □cite□turn1search11□
- NMAD is a robust DEM accuracy metric and should be reported if you evaluate against a reference DEM. □cite□turn7view0□turn7view1□

DL method design (inputs, conditioning, outputs)

Inputs (per pair edge)

1. Primary representation choices (pick one and justify): 1.1) Complex interferogram (I_) as two channels ((\Re I, \Im I)).
   1.2) Wrapped phase (\phi_=\arg(I_)) plus amplitude (|S_i|,|S_j|).
   1.3) Optional baseline coherence channel (\gamma_). □cite□turn10search2□turn0search0□

Conditioning metadata

2. Condition the network on:
   - (\Delta t), (\Delta\theta_\text), (\Delta\theta_\text),
   - mode, look_direction, flight_direction,
   - estimated (B_\perp),
   - SNR proxies (NESZ / amplitude stability). □cite□turn9view0□turn3view1□turn2view2□

Outputs

3. Output: 3.1) Denoised complex interferogram (\widehat*).*
   3.2) Per-pixel uncertainty map (\widehat{\sigma}^2(p)) (predict log-variance).
   This aligns with the need for downstream weighting and robustness. □cite□turn1view0□turn16search13□

Architecture suggestions (CNNs, equivariance, FiLM, graph conditioning)

1. **U-Net on complex-as-2-channels** + FiLM conditioning:
   - simple, robust, fastest to implement; strong baseline in phase filtering literature. □cite□turn10search2□turn0search0□
2. **Complex U-Net**:
   - more physically aligned, but more engineering risk.
3. **Graph-conditioned FiLM**:
   - compute node embeddings (collect-level) and feed edge embedding into FiLM for each pair; explicitly stack-aware and novel. This is the most contest-aligned "temporal storytelling" DL contribution.

Self-supervised and physics-consistency losses (explicit formulas required)

Let the model be (f_\theta(\cdot)\to(\widehat,\log\widehat{\sigma}^2)).

Noise2Noise / multilook self-supervision

Construct two noisy views (I^{(a)}*), (I^{(b)}*) using:

- sub-look splits / aperture splits, or
- stochastic multilook windowing.

Noise2Noise principle: learn from noisy–noisy pairs. □cite□turn15search2□turn10search7□

Loss: [\mathcal*{\text} = \mathbb\left[| \widehat*-I^{(b)}_|_1\right]]

Uncertainty (heteroscedastic NLL): [\mathcal*{\text} = \sum_p \left(\frac*{|\widehat*(p)-I^{(b)}*(p)|2^2}*{\widehat*{\sigma}^2(p)}+\log \widehat*{\sigma}^2*(p)\right)]

Closure-consistency triplet loss

Triplet closure is used to detect unwrapping errors and consistency. □cite□turn16search0□turn16search8□

[c_(p)=\mathrm\left(\widehat{\phi}*(p)+\widehat*{\phi}*(p)-\widehat{\phi}*(p)\right)] [\mathcal*{\text}=\sum_p w(p)\left(1-\cos(c_(p))\right)]

where (w(p)) is derived from uncertainty/coherence.

Temporal-consistency loss (stack-inversion residual)

[x^=\arg\min_x |W(Ax-\widehat*{\phi})|2^2] [\mathcal*{\text}=|W(Ax*^-\widehat{\phi})|_2^2] motivated by the emphasis on phase consistency in time-series InSAR processing. □cite□turn16search1□turn16search21□

Spectral/fringe-preservation loss

Prevent fringe destruction: [\mathcal*{\nabla}=|\nabla \widehat*{\phi}-\nabla \phi_|_1] This is aligned with phase filtering work that emphasizes preserving informative structures for unwrapping. □cite□turn10search2□turn0search6□

Overall: [\mathcal=\lambda_1\mathcal*{\text}+\lambda_2\mathcal*{\text}+\lambda_3\mathcal*{\text}+\lambda_4\mathcal*{\nabla}]

Training regimen (AOI split, augmentation, batch, checkpoints)

AOI-based splits (required by request)

1. Split by AOI: 1.1) Train AOIs: 60–70%
   1.2) Validation AOIs: 10–20%
   1.3) Test AOIs: 20–30%
   This avoids geographic leakage and supports generalization claims consistent with dataset intent. □cite□turn9view0□turn19view0□

Augmentation

2. Physically safe augmentations: 2.1) rotations by 90° multiples, flips,
   2.2) global phase offset (\phi\mapsto\mathrm(\phi+\Delta)),
   2.3) amplitude scaling in log space,
   2.4) random multilook window jitter for Noise2Noise. □cite□turn15search2□

Batch strategy

3. Two-level sampling: 3.1) sample AOI → sample edge → sample patch.
   3.2) upsample rare modes/high incidence differences to avoid bias.

Checkpointing

4. Save: 4.1) best-by-validation closure error,
   4.2) best-by-validation unwrap success,
   4.3) final epoch.
   Store `config.yaml`, git hash, and dataset manifest for each run. □cite□turn1view0□

Compute options table (requested)

Option	GPUs	RAM	Suitable for	Notes
Single GPU workstation	1×24 GB	128 GB	Minimal to functional	run fewer AOIs; smaller batches
Dual GPU workstation	2×48 GB	256 GB	Competitive	strong ablations; multi-AOI
Multi-GPU server	4–8×80 GB	512 GB+	Ideal	large-scale claims + hyperparam sweeps

Evaluation plan, reproducibility checklist, and paper vs repo packaging

Evaluation plan (metrics tables + baselines + ablations)

Baselines to compare (required list)

1. SNAP-style baseline for coherence/ifg (where compatible). □cite□turn0search2□
2. Multilooking + Goldstein filtering. □cite□turn16search2□turn16search6□
3. NL-InSAR. □cite□turn16search7□
4. BM3D-like denoising baseline. □cite□turn15search17□turn15search13□
5. Your DL model variants: N2N-only, +closure, +temporal, +uncertainty.

Ablation studies (minimum set)

- Remove metadata conditioning (FiLM off).
- Remove closure loss.
- Remove temporal loss.
- Remove uncertainty output; replace with fixed weighting.

Tables to compute (suggested)

1. Main metrics table per AOI and aggregated:

- closure median and quantiles
- unwrap success rate
- usable pairs %
- temporal residual norm
- DEM NMAD where applicable

2. Pair-selection table:

- edges proposed by heuristic vs predictor
- fraction usable and compute cost.

Reproducibility & code release checklist (required by contest)

The contest explicitly requires public code; disqualifying failures include missing code or unclear reproducibility. □cite□turn1view0□turn19view0□

1. `README.md` 1.1) one-line purpose
   1.2) environment creation
   1.3) one-command reproduction.

2. `REPRODUCIBILITY.md` 2.1) STAC root URL and contest collection ID used. □cite□turn5view0□turn6view0□
   2.2) exact download manifest and checksums.
   2.3) fixed random seeds; deterministic settings.

3. `configs/` 3.1) default config for baseline and DL runs.
   3.2) AOI split definitions.

4. `scripts/` 4.1) `download_subset.py` (STAC crawl → manifest → download).
   4.2) `preprocess_pairs.py` (coreg → ifg → coherence).
   4.3) `unwrap_snaphu.py`.

5. `eval/` 5.1) compute all metrics and generate figures used in the paper.

What goes in the 4-page paper vs repo (explicit)

4-page paper (must include):

1. Problem framing: geometry-diverse temporal InSAR stacks and why pair-graph + closure matters. □cite□turn1view0□turn16search0□
2. Method diagram and key equations (pair scoring; DL loss; uncertainty integration).
3. Main results table (metrics) + 2–3 striking visual comparisons.
4. Ablation table (small).
5. Short scalability discussion.

Supplementary repo (should include):

- full scripts, manifests, configs, trained weights
- extended AOI results and extra visuals
- any interactive exploration notebook ("temporal storytelling").

Mermaid charts requested: timeline and deliverables table

Deliverables timeline table (requested)

Date window (2026)	Deliverable	Why it matters
Mar 05–Mar 10	STAC crawl + metadata parquet + subset download	unlock experimentation quickly
Mar 10–Mar 18	Baseline InSAR products + unwrapping + closure metrics	establishes credible baseline
Mar 18–Mar 28	DL training (N2N + uncertainty) + integrate	first measurable improvements
Mar 28–Apr 02	Add closure + temporal losses, run ablations	turns into contest-level science
Apr 02–Apr 06	Final figures + 4-page paper + repo cleanup	submission quality wins committees

Mermaid timeline (requested)

gantt title DFC26 Submission Plan (as of 2026-03-05) dateFormat YYYY-MM-DD axisFormat %b %d section Data STAC crawl + local index :a1, 2026-03-05, 5d Subset download (AOIs) :a2, 2026-03-06, 8d section Baselines Coreg + interferograms :b1, 2026-03-10, 6d Goldstein/NL-InSAR/BM3D nets :b2, 2026-03-12, 6d SNAPHU unwrapping + closure :b3, 2026-03-14, 6d section Deep Learning N2N dataset creation :c1, 2026-03-16, 4d Train + uncertainty head :c2, 2026-03-18, 8d Add closure + temporal losses :c3, 2026-03-24, 8d section Paper + Release Metrics tables + ablations :d1, 2026-03-30, 5d Manuscript + visuals + repo :d2, 2026-04-02, 4d Submit :milestone, 2026-04-06, 0d

Full references list

The references below are prioritized to match your requested categories (contest/data → core InSAR stacks → closure/unwrapping → filtering baselines → DL methods across CNN/GAN/transformer/diffusion → DEM metrics).

Contest, data access, formats, STAC

- 2026 Data Fusion Contest page (rules, schedule, evaluation criteria, dataset size, access, CSV headers). □cite□turn19view0□turn9view0□turn1view0□
- AWS Registry: Capella Space SAR Open Dataset (no-auth S3 access; region; license). □cite□turn13view0□
- Capella Open Data STAC root `catalog.json` and contest child link. □cite□turn5view0□turn6view0□
- Capella SAR Products Format Specification v1.8 (SLC complex format; calibration; scale_factor; TIFF+JSON bundle). □cite□turn3view0□turn3view1□turn3view2□
- capella-reader GitHub repository (metadata parsing; ISCE3 adapter). □cite□turn2view2□
- STAC spec repository (static catalogs) and OGC STAC API description. □cite□turn0search5□turn0search17□turn0search1□

Core InSAR stacks and time series

- Crosetto et al. (2016) PSI review (principles and time-series monitoring context). □cite□turn0search19□
- Berardino et al. (SBAS) overview reference (small baseline differential interferograms). □cite□turn0search18□
- Phase consistency emphasis in SBAS-style processing (recent). □cite□turn16search1□

Closure, temporal consistency, unwrapping

- Triplet closure importance for identifying unwrapping errors and quality control. □cite□turn16search0□turn16search8□
- Closure phase bias in multilooked InSAR (systematic considerations). □cite□turn16search13□turn16search5□
- SNAPHU software page (statistical-cost, network-flow MAP unwrapping). □cite□turn15search8□turn15search4□

Classical filtering baselines

- Goldstein & Werner interferogram filtering (adaptive filter improves phase noise and unwrapping). □cite□turn16search2□turn16search6□
- NL-InSAR nonlocal interferogram estimation. □cite□turn16search7□
- BM3D denoising (as BM3D-like baseline). □cite□turn15search17□turn15search13□

DL methods: CNNs, GANs, transformers, diffusion

- DeepInSAR (CNN for phase filtering + coherence estimation). □cite□turn0search0□turn0search16□

- Murdaca et al. DL phase filtering framework (encoder–decoder; coherence/phase prediction). □cite□turn10search2□turn0search20□
- Φ-Net / Phi-Net (DLR) for joint phase and coherence estimation. □cite□turn0search12□turn0search8□
- U-Net unwrapping/denoising approaches (deep unwrapping trend). □cite□turn10search0□turn10search1□
- ResUCTransNet (transformer elements for unwrapping). □cite□turn0search14□
- Two-stage GAN for interferometric phase reconstruction in decorrelated areas. □cite□turn0search5□
- SAR despeckling with diffusion models (SAR-DDPM) and diffusion+transformer variants. □cite□turn0search3□turn0search7□
- Conditional diffusion for InSAR unwrapping (recent preprint trend; treat as exploratory). □cite□turn1search7□
- SAR2SAR (self-supervised multi-temporal SAR despeckling concept). □cite□turn10search7□
- VAE-related SAR compression/despeckling work (VAE is relevant as a representation baseline). □cite□turn1search2□

DEM accuracy metric

- Höhle & Höhle (2009): NMAD definition and robust DEM accuracy assessment. □cite□turn7view0□turn7view1□